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Thermoelectric Performance of Poly(3,4-ethylenedioxythiophene):Poly(styrenesulfonate) *

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Thermoelectric (TE) performances are systematically investigated for the pellets of poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) with different organic additives and heating process as organic TE materials. The electrical conductivity, Seebeck coefficient and thermal conductivity versus temperature are determined, respectively. It is found that there is no distinct change for the Seebeck coefficient among each sample with the additions of dimethyl sulfoxide and ethylene glycol. The thermal conductivity measured in a wide range of temperature indicates that the PEDOT:PSS pellet have an extremely low value. The highest figure of merit ($Z_T = 1.75 \times 10^{-3}$) is observed at 270 K among the PEDOT:PSS pellets.

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Thermoelectric (TE) materials have attracted significant attention in current intense research. The performance of TE materials depends on the dimensionless figure of merit, Z_T , given by the following equation

$$Z_T = \frac{\sigma \cdot S^2}{\kappa} T, \quad (1)$$

where σ , S , κ and T are the electrical conductivity, Seebeck coefficient, total thermal conductivity and absolute temperature, respectively. In addition, σS^2 is known as the power factor. Presently, most studies on the high-efficiency TE materials mainly focus on the field of inorganic semiconductor, and the best TE materials commercially available have $Z_T \approx 0.9$.^[1,2] While only a few were investigated for TE materials of about organic conducting polymers (CPs), mainly including polyacetylene,^[3] polyaniline,^[4] polythiophene,^[5] polypyrrole,^[6] and their derivatives,^[7] etc. Compared to inorganic solids, apart from the very low thermal conductivity, conducting or semiconducting polymers with desirable electronic band structure favouring high power factor, in principle, can be designed and may be synthesized.^[8]

Among numerous CPs, poly(3,4-ethylenedioxythiophene) (PEDOT), one of polythiophene derivatives, is well known and considered as the most remarkable CP, which has been attracting particular interest.^[9,10] Furthermore, PEDOT has several advantages, especially the high electrical conductivity (about 400–600 Scm⁻¹) and good environmental stability in conducting state at ambient temperature.^[11,12] Another

point is crucial that PEDOT can be processed in aqueous solution with poly(styrenesulfonate) (PSS) by appropriate ratio and a homogeneously dispersed polymer complex (PEDOT:PSS) is formed. Based on these characteristics, PEDOT:PSS (Fig. 1(a)) has been widely applied in the design of antistatic coatings, electrode materials, field-effect transistors, organic light-emitting diodes (OLED) and photovoltaic cells.^[12–14] Additionally, it is fascinating that the electrical conductivity can be extremely enhanced by the addition of various organic solvents.^[14–16] Thus, PEDOT:PSS would be the promising candidate material for TE owing to its advantages of high electrical conductivity, low thermal conductivity, light-weight, flexibility, excellent thermal stability.^[7–10]

The PEDOT:PSS solution (baytron P) was used directly. Here 5% and 10% organic solvents of dimethyl sulfoxide (DMSO) and ethylene glycol (EG) (Fig. 1(b)) by the volume were mixed with the PEDOT:PSS solution. It should be noted that sample A does not contain any additives and is dried in vacuum at 40°C. The samples B–I contained the amount of additives: (B) 5% DMSO, (C) 10% DMSO, (D) 5% DMSO, (E) 10% DMSO, (F) 5% EG, (G) 10% EG, (H) 5% EG, (I) 10% EG, respectively. The mixed solutions were stirred continuously for 24 h at room temperature, and then were dried in vacuum at 40°C. Finally, the PEDOT:PSS pellets A–I were compressed for the measurements under a pressure of 20 MPa. In addition, samples D, E, H, I were baked at 150°C for 30 min before pressed.

The electrical conductivity of all the samples was

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measured using the four point probe technique as a function of absolute temperature, as shown in Fig. 2. It is clear that the influence of additives and heating process for the conductivity does not show well-regulated law. However, sample A with no additive has the lowest value in comparison with others. These results indicate that the organic additives play a key role for the conductivity enhancement of PEDOT:PSS, which is similar to the results reported previously.^[13,16,17]

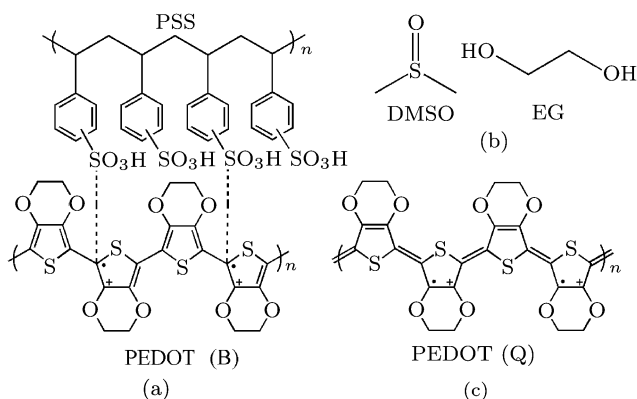


Fig. 1. Chemical structures of (a) PEDOT:PSS and benzoid (B) type of PEDOT, (b) DMSO and EG, (c) quinoid (Q) type of PEDOT. The dots and pluses represent the unpaired electron and positive charge on the PEDOT chain, respectively.

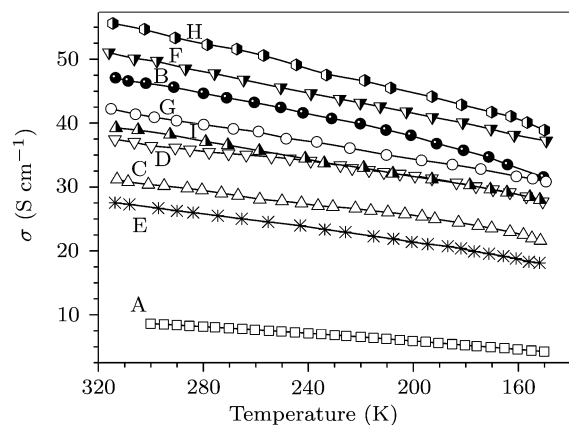


Fig. 2. Electrical conductivity of PEDOT:PSS pellets A–I versus temperature.

It is important to realize that structures play a dominant role in determining the physical properties of CPs. The electronic properties are totally different from conventional semiconductors. The electron–hole interaction leads to strongly bonded excitons and greatly affects the transport properties

in CPs. The charge carrier transport mechanism in the case of PEDOT:PSS can be generally explained within the framework of the charging-energy limited tunnelling model, originally proposed for highly disordered CPs.^[18] The doping process (by adding EG, DMSO) results in charges along the conjugated backbone of the PEDOT which makes the flow of current.^[19] Currently, the enhancement of mechanism for electrical conductivity with additives still exists a lot of controversies. We relatively approve that the resonant structure of PEDOT chain changes from a benzoid (B) to a quinoid (Q) structure (Fig. 1(c)) and the interaction between additives and PEDOT chains. This means that great change from a coil conformation to a linear or expanded-coil conformation has dramatically taken place in the structure of PEDOT, before and after processing,^[15] which is in agreement with our results of scanning electron microscopy (SEM). Moreover, the PEDOT:PSS pellets with additives show more smooth surface in comparison with the sample without additives in these SEM images (Fig. 3). Therefore, it is understandable that the change of structure in PEDOT leads to the enhancement of electrical conductivity after adding DMSO or EG.

On the other hand, the Seebeck coefficient of materials, which is a intrinsic property, is defined as

$$S = \frac{\Delta V}{\Delta T}, \quad (2)$$

where ΔV and ΔT are the voltage drop across the material and the temperature gradient along the voltage drop. It was determined by establishing a temperature gradient of $\Delta T = 5 \text{ K}$ and measuring the output voltage between both ends of sample, ΔV . Figure 4 shows the temperature-dependent Seebeck coefficient of each sample. The values of Seebeck coefficient are varying between 8 and $15 \mu\text{V K}^{-1}$ from 150 to 300 K. In sharp contrary to electrical conductivity, sample A with no additives keeps higher result than others. Simultaneously, there is no distinct change for the values among samples. The reported value was of about $10 \mu\text{V K}^{-1}$ at room temperature,^[17] which was basically in accordance with our results around $13 \mu\text{V K}^{-1}$. In terms of previous investigations, a series of polycarbazole and its derivatives possessed high Seebeck coefficient around $200\text{--}250 \mu\text{V K}^{-1}$ and the highest value could reach up to $600 \mu\text{V K}^{-1}$ at low doping level.^[7,20] It is significant and expected that the Seebeck coefficient of copolymers of PEDOT with other CPs can reach to $200 \mu\text{V K}^{-1}$.

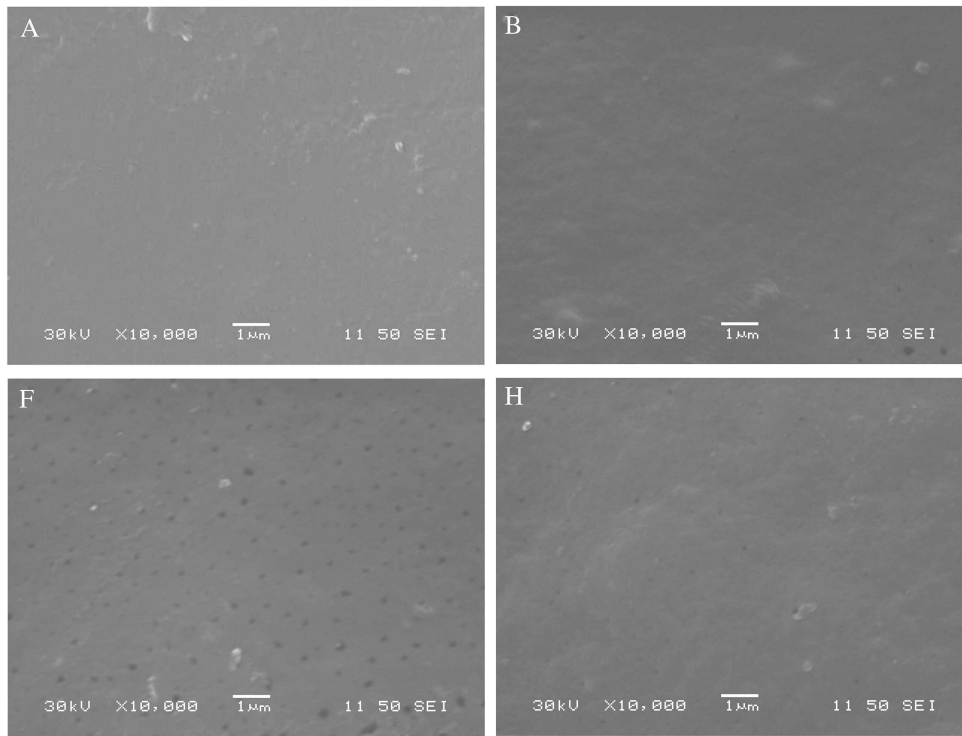


Fig. 3. SEM images of samples A, B, F and H.

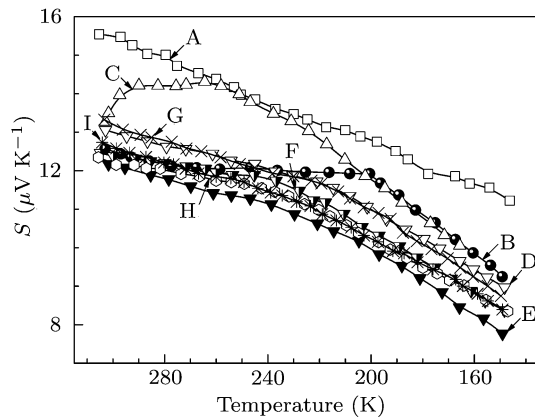


Fig. 4. Seebeck coefficient of PEDOT:PSS pellets A–I versus temperature.

The Seebeck coefficient is an intrinsic electronic transport property and is also of practical importance. It is affected by charge carrier interactions with one another, with phonons and the local magnetic moments of magnetic solids.^[21] Generally, the increase in carrier concentration causes the proportional increase in electrical conductivity but the logarithmic decrease in Seebeck coefficient in conventional semiconductors or metals.^[22,23] However, the Seebeck coefficient hardly takes place to change too greatly with the enhancement of electrical conductivity, which is different from the conventional semiconductors. The possible reason for this phenomenon is that the more increase of carrier concentration than that in car-

rier mobility results in the enhancement of electrical conductivity with retaining stable Seebeck coefficient. Moreover, the electronic structure of PEDOT:PSS is also likely to be a main factor. Therefore, the phenomenon that Seebeck coefficient is constantly basic to keep accompanied by the enhancement of electrical conductivity can not be explained in detail now.

Figure 5 presents the trend of thermal conductivity of sample E as a function of temperature. The thermal conductivity can be calculated by

$$\kappa = \frac{Q \cdot L}{A \cdot \Delta T}, \quad (3)$$

where Q , L , A and ΔT are the power supplied, the distance of sample, the cross-sectional area of sample and the temperature gradient along the voltage drop, respectively. It is significant that PEDOT:PSS pellets inherently possess a low thermal conductivity about $0.17 \text{ W m}^{-1} \text{ K}^{-1}$ under ambient condition over conventional TE materials (about $2 \text{ W m}^{-1} \text{ K}^{-1}$).^[24,25] The thermal conductivity was determined by making use of a heat and sink steady state method by applying a smooth temperature difference, and supplying quantity of heat at one end of the sample while the other end is maintained at a constant temperature. The temperature difference ΔT and the fixed distance L were measured. It is noticeable that an accuracy about $1 \times 10^{-2} \text{ K}$ was maintained between the two ends of the sample before measurement. In order to avoid the thermal losses, special attention was taken in the process of measurement.

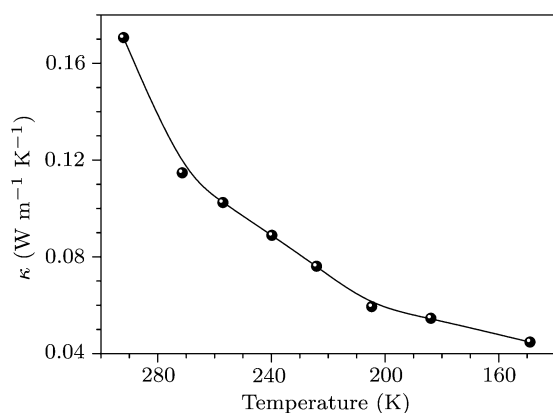


Fig. 5. Thermal conductivity of PEDOT:PSS pellet E versus temperature.

According to the above experimental results, Fig. 6 compares the Z_T values of each sample as a function of absolute temperature. It should be noted here that the thermal conductivity of all the samples is based on Fig. 4, because there are no inorganic dopants in these samples. Vivid diversity can be observed for each sample. Samples B–I possess of higher Z_T values than sample A. At ambient temperature, the Z_T values of PEDOT:PSS pellets are higher than polyaniline films ($Z_T \sim 1 \times 10^{-4}$) by one order of magnitude.^[26] Moreover, the Z_T value attains the maximum of 1.75×10^{-3} near 270 K with potential probability of extensive enhancement, which is a comparatively ideal value in the field of CPs.

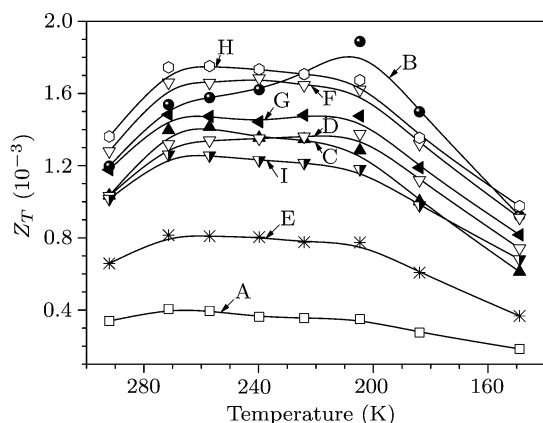


Fig. 6. Temperature dependence on thermoelectric figure of merit Z_T of PEDOT:PSS pellets A–I.

In summary, we have systematically investigated the thermoelectric performance of PEDOT:PSS pellets with different organic additives such as DMSO, EG and with a comparatively good Z_T values of

10^{-3} in the field of CPs. Considering the high Seebeck coefficient as polycarbazoles together with the optimal electrical conductivity of PEDOT, it can be expected that conducting copolymers of 3,4-ethylenedioxythiophene (EDOT) and other monomers with higher Z_T values by one or two orders of magnitude will be produced, which would be fascinating and challenging in both academic and industrial as organic TE materials.

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